Does the gender gap in high income survive matching on job and education? Evidence from the 1994 US Adult census data

Group name	C5XW20
Dataset	kaggle.com/datasets/mlbysoham/adult-dataset (UCI Adult, archive.ics.uci.edu/dataset/2/adult)
Code	github.com/durgesh-k-sharma/c5xw20-midterm (scripts, README, pre-registration, outputs)

#	Name	Roll no.	Institute email	Contribution
1	Aryan Bundela	2023141	aryan23141@iiitd.ac.in	Dataset, size check (1); repo, README
2	Durgesh Kumar Sharma	2023214	durgesh23214@iiitd.ac.in	Preparation, before and after (2, 3)
3	Shourya Sharma	2020129	shourya20129@iiitd.ac.in	EDA, problem statements (4); Test 1
4	Arpit Dabas	2023131	arpit23131@iiitd.ac.in	Tests 2 to 4 (5); limitations (6)

1. Dataset

Source: Kaggle, "Adult Dataset" (user mlbysoham). It is the file `adult.data`, the training part of the UCI Adult (Census Income) data by Becker and Kohavi (1996), CC BY 4.0. UCI says it was extracted from the 1994 US Census database. One row is one person in a survey sample. We use 32,561 of the 48,842 UCI rows (66.7%).

What is raw about it: the file has no header row. Missing values are written as `?`: 4,262 cells in 2,399 rows, in three columns. 24 rows are exact duplicates. `education` and `education-num` carry the same information (16 levels each, one to one). `capital-gain`, `hours-per-week` and `age` are capped at 99,999, 99 and 90. Money is in 1994 dollars. UCI already removed some records (for example age 16 or below), so the data is only lightly dirty, and we say so rather than claim it is raw.

	Raw file	After encoding
Instances (rows)	32,561	32,537
Attributes (columns)	15	108
Instances $\times$ attributes	488,415	3,513,996

Size check: $32{,}537 \times 108 = 3{,}513{,}996 \geq 3{,}000{,}000$, so the requirement is met. The 108 columns are 6 numeric columns plus 102 one-hot columns (the 8 nominal columns have $9 + 16 + 7 + 15 + 6 + 5 + 2 + 42 = 102$ levels). The response `income` is not counted. The margin is thin: if we had dropped the rows with unknown values and the repeated `education` column, we would have $30{,}139 \times 88 = 2{,}652{,}232$, which fails. So we keep both choices (see Section 2).

Key features

The full data dictionary (type, unit, range, missingness for all 15 columns) is in the repo file `outputs/data_dictionary.csv`.

Feature	Type	Real-world meaning, range, missing
`income`	nominal	response: yearly income >50K or not; 24.1% are >50K
`sex`	nominal	Female or Male; none missing
`age`	discrete	years, 17 to 90 (90 is a cap)
`education`	nominal	highest level, 16 levels; `education-num` codes it 1 to 16
`occupation`	nominal	job type, 14 levels and `?`; 1,843 missing
`workclass`	nominal	employer type, 8 levels and `?`; 1,836 missing
`hours-per-week`	discrete	usual weekly hours, 1 to 99
`capital-gain`	continuous	yearly gains in 1994 USD, 0 to 99,999
`fnlwgt`	continuous	census weight: people the row stands for

2. Data preparation

Issue found	What we did	Why, and not the alternative
No header; `?` for missing	added names; read `?` as a level called `Unknown`	1,836 rows have `workclass` and `occupation` both `?`: one group of people with no job information, not random gaps. In the exploration half only 10.8% of them earn >50K, against 24.9% for the rest, so the group carries information. A mode fill would invent jobs. Dropping rows loses 7.4% and fails the size check.
24 exact duplicate rows	dropped	they match on all 15 columns including the weight `fnlwgt`, so they are copies. 3,465 rows match when `fnlwgt` is ignored; we keep them, because they are different people with similar profiles.
`education` repeats `education-num`	kept both; tests use 4 education bands	dropping one column would break the size check. Bands (below HS, HS grad, some college, Bachelors or more) give strata big enough to test.
Capped values: 159 gains at 99,999, 85 at 99 hours, 43 at age 90	kept; three flag columns added	a cap is how the survey recorded a large value, not an error. Replacing it would invent a number. Flags let us leave them out.
IQR rule flags 9,002 hours rows (28%) and 2,712 gains	kept all rows; used rank tests and robust standard errors	most people work exactly 40 hours or have zero gain, so the IQR is tiny and the rule flags normal values. Deleting them would remove real people.
`fnlwgt` survey weight	kept, not used in the main tests	course tests treat each row as one person. One weighted check for Test 1 shows if this matters.
8 nominal columns	one-hot encoded; `education-num` kept as a number	no order exists between nominal levels, so integer codes would invent one.
Split	50/50, stratified on `income`, seed 558, after removing duplicates	EDA used the exploration half (16,268 rows); tests used the confirmation half (16,269 rows). The >50K share is 24.09% and 24.09%.

3. What changed after preparation

Measure	Before	After
Rows	32,561	32,537
Attributes	15	108
Rows × attributes	488,415	3,513,996
Duplicate rows	24	0
`?` cells	4,262 (2,399 rows)	0; 4,261 cells (2,398 rows, 7.4%) are now the level `Unknown`
Capped values	not marked	flagged: 159 + 85 + 43

Encoding took the attributes from 15 to 108, which clears the size requirement. We lost 24 rows (0.07%) and nothing else. We gained a clean `Unknown` level and bands for education. We did not remove any outlier, so every extreme value is still in the data. One duplicate row held a `?`, so the `Unknown` count (4,261) is one lower than the raw `?` count (4,262).

4. Basic questions and problem statement

All figures here use the exploration half only (16,268 rows). The confirmation half was not opened until the tests.

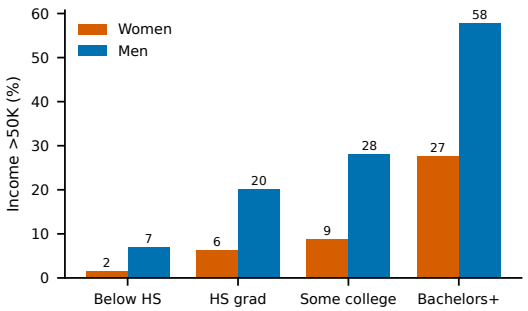


Figure 1: Men lead women in every education band, by 5.3 to 30.3 points; the gap grows with education.

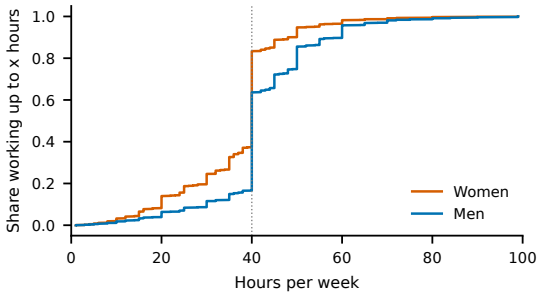


Figure 2: 37.4% of women work under 40 hours a week against 16.6% of men; mean hours are 36.6 and 42.5.

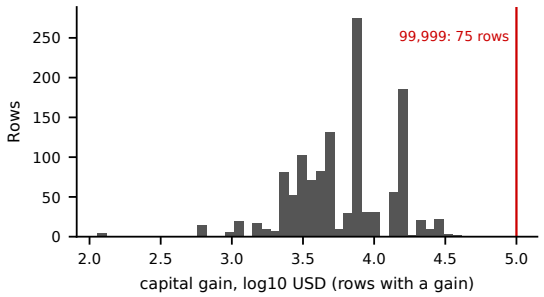


Figure 3: Capital gain is rare (8.3% of rows) and capped: all 75 rows at 99,999 earn >50K, so the cap marks the response.

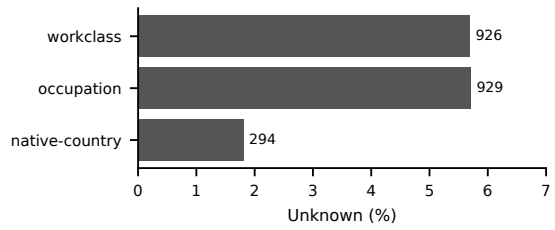


Figure 4: Unknown values sit in three columns; 926 rows (5.7%) lack both `workclass` and `occupation`.

Basic questions

1. Do men earn more than women? Yes in this half: 30.5% of men and 11.2% of women are above 50K.
2. Does more education mean higher income? Yes: the >50K share goes from 5.2% (below HS) to 49.0% (Bachelors or more); $r = 0.34$ with `education-num`.
3. Does income rise with age? Weakly: $r = 0.24$.

Our problem statements

P1 (primary). Among people with the same occupation and education band, are women less likely to earn above 50K than men?

P2. Is the sex gap in weekly hours different for married (civilian spouse present) people than for never-married people?

P3. Do people with a capital gain work different hours from people with none?

P4. Is income independent of the employer type (`workclass`)?

How they are novel: the basic questions compare two raw groups. P1 asks if the raw gap survives matching on job and education. In the exploration half it does: the matched odds ratio is 0.25 (48 usable strata of 55), almost the same as the raw one, 0.29. P2 asks about a *difference of two gaps*, which no single row records. P3 and P4 are closer to the basic questions. We keep them because they stress other tests: P3 has 46% of women and 47% of men at exactly 40 hours, and P4 has levels with a handful of rows.

Cause and effect: P1 sounds causal, but it is not. Matching on two variables leaves out age, hours, industry and more. Occupation may also be a path through which a gap arises, and matching removes it. A causal claim would need a design with the same people compared with only the sex changed, or a full model of how jobs are assigned. We only claim an association.

Disclosure: before the split, we saw two whole-data summaries (income share and mean hours by sex). `PREREGISTRATION.md`, committed as 8cfa81f before any script ran, lists this and fixes every test below.

5. Statistical inferences

Rows tested: the confirmation half, 16,269 people. Test 1 uses people with a known occupation (914 unknown rows out) and only strata that contain both sexes, which leaves 15,323 people (4,934 women, 10,389 men) in 51 of 54 strata (32 rows lost). Test 2 uses 12,803 people. Test 3 uses all. Test 4 drops 910 unknown rows. We use $\alpha = 0.05$ for every test, with a Holm correction across the four p-values. "Matters" means the whole 95% interval lies beyond the smallest difference; "does not matter" means it lies inside; otherwise "inconclusive".

Test 1 (P1): do women have lower odds of income above 50K within the same job and education band?

a) Hypotheses, written before running the test

H_0	$\mathrm{OR} \geq 1$
H_1	$\mathrm{OR} < 1$
Significance level	$\alpha = 0.05$
Side	one-sided (left)
Test	Cochran-Mantel-Haenszel (CMH)
Smallest difference that matters	$\mathrm{OR} = 0.80$

OR: common odds ratio, women over men, across the occupation $\times$ education-band strata. One-sided, because P1 says women are less likely. An OR of 0.80 means 20% lower odds, the smallest gap we would call worth noting.

b) Other tests we considered

A yes-or-no outcome, a yes-or-no exposure and many strata call for CMH: it combines one 2×2 table per stratum.

Also considered	Why we did not use it
Plain χ^2, one table	ignores the strata, so a mix of jobs could fake or hide a gap. Run anyway: $\chi^2 = 739.5$, $p = 7.9 \times 10^{-163}$, crude OR 0.273 (0.248 to 0.301). Same decision, but not matched.
Logistic regression with stratum dummies	also valid, and gives OR 0.253 (0.227 to 0.282), $z = -24.6$; it must drop strata with a constant outcome (15,267 rows, 47 strata) and fits a bigger model. CMH needs no model and is simpler.
Fisher exact test per stratum	counts are large, and 51 separate tests would need their own correction.
Breslow-Day test	not a test of P1: it asks if the OR is equal across strata. Tarone-adjusted $\chi^2 = 66.9$ (46 d.f.), $p = 0.024$: mild differences, and 42 of 46 strata with a defined OR point below 1.
Weighted logistic (check)	with `fnlwgt` weights and robust errors: OR 0.258 (0.231 to 0.288). Weights do not change the story.

c) Result

Estimate	$\widehat{\mathrm{OR}} = 0.253$ (women 11.2%, men 31.5% above 50K)
95% confidence interval	0.227 to 0.283 (Robins-Breslow-Greenland)
Test statistic (d.f.)	$\chi^2_{\mathrm{CMH}} = 646.4$ (1); $z = -25.4$
p-value (Holm-adjusted)	6.7×10^{-143} (2.7×10^{-142})
Decision	reject H_0; it matters

Within the same occupation and education band, women have about three quarters lower odds of earning above 50K (a pooled risk ratio of 0.39). The whole interval is below 0.80, so the gap matters. Matching did not shrink the gap: the crude OR was 0.273, the matched OR is 0.253. Job type and education level, as we measure them, do not explain it.

Test 2 (P2): is the sex gap in hours different for married and never-married people?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$D = 0$
$\boldsymbol{H_1}$	$D \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	t-test on the interaction in OLS, HC3 errors
Smallest difference that matters	2 hours per week

D is the men-minus-women gap in mean weekly hours among married-civ-spouse people, minus the same gap among never-married people. Two-sided, because we did not know the direction in advance. Two hours is about one working day a month.

b) Other tests we considered

We compare a difference of two gaps across two factors, so we need an interaction test. We use OLS with robust errors. Cell spreads run from 11.7 to 12.4 hours, and a Brown-Forsythe test rejects equal spreads ($F = 12.1$, $p = 6.5 \times 10^{-8}$), so we do not assume them, though the spreads are close.

Also considered	Why we did not use it
Classic OLS t-test	assumes equal variances. Run anyway: $t = 6.85$, $p = 7.5 \times 10^{-12}$, same decision.
Permutation test	makes no normality claim: sex shuffled within marital group 10,000 times, $p < 10^{-4}$ (no shuffle reached $\|D\|$; the null SD is 0.56 against $D = 3.79$). Same decision.
Two Welch t-tests	give a gap of 7.65 h for married (6.78 to 8.52) and 3.86 h for never married (3.20 to 4.53), but no test of their difference by themselves. A Wald z on the two gaps gives $z = 6.79$, $p = 1.1 \times 10^{-11}$, the same as the OLS result.
Mann-Whitney, paired t-test	Mann-Whitney compares two groups, not a difference of two gaps; no person is in two groups.

c) Result

Estimate	$\hat{D} = 3.79$ h (gap 7.65 h married, 3.86 h never married)
95% confidence interval	2.70 to 4.88
Test statistic (d.f.)	$t = 6.79$ (12,799)
p-value (Holm-adjusted)	1.2×10^{-11} (1.2×10^{-11})
Decision	reject H_0; it matters

The sex gap in hours is about 3.79 hours larger among married people. The whole interval lies above 2 hours. Mean hours: married men 44.0, married women 36.3; never-married men 38.6, women 34.7. With a standard error of 0.56 h, a test of this size can detect a difference of about 1.6 h (2.8 standard errors, 80% power), below our 2-hour line. This is a pattern, not a cause: we did not match the marital groups on age.

Test 3 (P3): do people with a capital gain work different hours?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	the two hours distributions are equal (no shift)
$\boldsymbol{H_1}$	they differ (a shift)
Significance level	$\alpha = 0.05$
Side	two-sided
Test	Mann-Whitney U
Smallest difference that matters	2 hours per week

Groups: people with a capital gain ($n = 1{,}362$) and with none ($n = 14{,}907$). The estimate is the Hodges-Lehmann shift, the median of all pairwise differences. Two-sided, because we had no direction.

b) Other tests we considered

Hours are whole numbers, skewed, and 41% and 47% of the two groups sit at exactly 40. A rank test needs no normality. The price is that it tests whether one group tends to have larger values, which is not the same as a gap in means.

Also considered	Why we did not use it
Welch t-test	valid at this size and answers a different question, a gap in means: 3.00 h (2.34 to 3.65), $t = 8.98$ (1,649), $p = 7.2 \times 10^{-19}$. Its interval is wholly above 2 h, so it matters.
Brunner-Munzel test	allows unequal spreads and is the safer rank test: $\|W\| = 10.97$, $p = 4.7 \times 10^{-27}$. Same decision.
Two-sample z-test	needs a known σ.
Median test	both medians are 40 h, so it has nothing to find.

c) Result

Estimate	Hodges-Lehmann shift 0.0 h (medians 40 and 40; means 43.1 and 40.1)
95% confidence interval	0.0 to 0.0 (bootstrap, 2,000 resamples)
Test statistic	$U = 11{,}868{,}086$, $z = 10.92$; ($n_1 = 1{,}362$, $n_2 = 14{,}907$)
p-value (Holm-adjusted)	8.8×10^{-28} (1.8×10^{-27})
Decision	reject H_0; on the Hodges-Lehmann estimate it does not matter

The two tests disagree about size, and the reason matters. The distributions differ (a random person with a gain works more hours than a random person without one with probability 0.585, counting ties as half). But the typical person in both groups works 40 hours, so the Hodges-Lehmann shift is 0.0 and its interval has zero width. The difference is in the tails: 42% of the gain group work over 40 hours against 28%, and 17% work under 40 against 25%. The pre-registered rule gives "does not matter"; the mean gap of 3.00 hours does matter. We report both and do not pick the one that suits us.

Test 4 (P4): is income independent of employer type?

a) Hypotheses, written before running the test

H_0	income (>50K or not) is independent of `workclass`
H_1	they are not independent
Significance level	$\alpha = 0.05$
Side	two-sided (a χ^2 test has no direction)
Test	Pearson χ^2 test of independence
Smallest difference that matters	Cramér's $V = 0.10$

`Unknown` rows are out. `Never-worked` and `Without-pay` merge into `No-pay` (10 people), because each is too small alone. $V = 0.10$ is the usual cut for a small association.

b) Other tests we considered

A table of two categorical variables (7×2) calls for a χ^2 test of independence. Its rule of thumb holds: 1 of 14 cells has an expected count under 5 (smallest 2.49), and none is under 1.

Also considered	Why we did not use it
G-test (likelihood ratio)	gives $G = 398.0$ (6 d.f.), $p = 7.6 \times 10^{-83}$; same decision, and no better at this size.
Monte Carlo χ^2	10,000 shuffles of income, $p < 10^{-4}$. Same decision; it does not lean on the large-sample rule for the `No-pay` row.
Fisher-Freeman-Halton exact	not feasible for 15,359 rows; the Monte Carlo test stands in for it.
Z-test per level	compares only two groups; it does not test all seven together.

c) Result

Estimate	Cramér's $V = 0.172$ (`Self-emp-inc` 58.4%, `Private` 22.0% above 50K)
95% confidence interval	0.154 to 0.190 (bootstrap, 2,000 resamples)
Test statistic (d.f.)	$\chi^2 = 452.3$ (6)
p-value (Holm-adjusted)	1.6×10^{-94} (4.7×10^{-94})
Decision	reject H_0; it matters

Income depends on employer type, and the whole interval is above 0.10. The self-employed with an incorporated business have the highest >50K share (58.4%); private employees have 22.0%.

Summary of the four tests

Test	Estimate (95% CI)	Line	Holm p	Verdict
1	OR 0.253 (0.227 to 0.283)	0.80	2.7×10^{-142}	matters
2	$\hat{D}$ 3.79 h (2.70 to 4.88)	2 h	1.2×10^{-11}	matters
3	HL 0.0 h (0.0 to 0.0); mean gap 3.00 h (2.34 to 3.65)	2 h	1.8×10^{-27}	HL: does not matter; mean: matters
4	V 0.172 (0.154 to 0.190)	0.10	4.7×10^{-94}	matters

All four reject H_0 after the Holm correction. Only Test 3 changes its verdict with the estimate we choose.

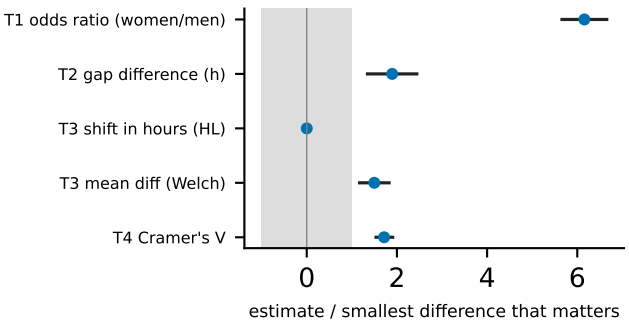


Figure 5: Each estimate in units of its smallest difference that matters. Grey is the zone where the difference does not matter; Test 3 lands in it only with the Hodges-Lehmann estimate.

6. Limitations

- **Old data.** The records are from 1994 and the 50K line is in 1994 dollars. The results describe that year.
- **Association only.** Matching leaves out age, hours and industry, and occupation may be a path of the gap, so P1 is not a proof of unequal pay.
- **Not fully blind.** We saw two whole-data summaries before the split (Section 4). The overall direction of P1, and of the hours gap behind P2, was already known.
- **Large n.** With about 16,000 rows, tiny gaps give tiny p-values, so we judge by the intervals against the thresholds. We chose those thresholds, and others could choose differently.
- **Test 3 depends on the estimate** (above), because of ties at 40 hours.
- **Independence.** The survey samples households, so spouses may both appear. The file has no household id to check this.
- **Weights.** Main tests ignore `fnlwgt`; one weighted check agrees (OR 0.258). Only the training half of UCI is used.
- **Small cells.** `No-pay` has 10 people; the comparison tests are not corrected for multiple testing, only the four main tests are.

References. Becker, B. and Kohavi, R. (1996). Adult. UCI Machine Learning Repository, archive.ics.uci.edu/dataset/2/adult. Kohavi, R. (1996). Scaling up the accuracy of naive-Bayes classifiers: a decision-tree hybrid. KDD-96.